

KUSHAGRA DUBEY

Backend Engineer · Node.js · NestJS · PostgreSQL · Docker · Kubernetes

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PROFESSIONAL SUMMARY

Backend Engineering Intern at FlyRank AI specialising in Node.js, NestJS, Express.js, PostgreSQL, Redis, Docker, and Kubernetes. Delivered a Usage Metering & Billing Engine and DevPulse — a full-stack analytics platform with LLM integration, cron automation, and Docker multi-container deployment. Experienced in REST API design, JWT authentication, microservices architecture, and CI/CD; seeking backend engineering roles building scalable, production-ready distributed systems.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python

Backend: Node.js, NestJS, Express.js, REST API Design, Microservices Architecture, JWT Authentication, Passport.js

Databases: PostgreSQL, MongoDB, Mongoose, Redis (caching · rate-limiting · pub/sub), Supabase

DevOps & Cloud: Docker, Docker Compose, Kubernetes (K8s), CI/CD Pipelines, Render, AWS (in-progress)

Frontend: React.js, Next.js, Tailwind CSS, Material UI, Redux, React Router, Axios, Chart.js, EJS

Tools: Git, GitHub, Swagger, OpenAPI, Zod, Helmet.js, Twilio API, OpenAI API, nodemailer

PROFESSIONAL EXPERIENCE

FlyRank AI · Backend Engineering Intern · Jun 2026 – Present

- Architected a production-grade Task Management REST API using Node.js, Express.js, and PostgreSQL with Swagger/OpenAPI documentation across 4+ sprint milestones; containerised with Docker Compose — multi-service orchestration, PostgreSQL health checks, and environment-variable secrets management
- Integrated Supabase JWT authentication (sign-up, sign-in, email provider); built Cheerio + Zod web scraper with Redis-backed caching, rate-limiting, and schema validation across 60+ records
- Capstone 1 — Usage Metering & Billing Engine: metered-billing backend using Node.js and PostgreSQL for per-user API consumption tracking, billing cycle computation, and invoice generation; Redis high-throughput counters for real-time metering
- Capstone 2 — DevPulse (Developer Productivity Tracker): full-stack platform — Node.js REST API, PostgreSQL, Redis caching, JWT auth, cron jobs, OpenAI/Claude LLM integration, PDF generation, nodemailer email delivery, Docker multi-container orchestration

PROJECTS

Stockify — Full-Stack Stock Trading Dashboard · React.js · Node.js · Express.js · REST API · JWT · Material UI

- Developed a full-stack stock trading simulation using React.js and Node.js/Express.js; implemented Bearer token JWT authentication with authenticated buy/sell endpoints (POST /addOrders, POST /sellOrder) and account management routes (GET /me, GET /allholdings) for user-specific holdings and balance tracking.
- Engineered real-time UI synchronisation post-transaction via browser custom events (stockify:data-updated, stockifyBalanceUpdated) and React state management; built responsive stock watchlist with case-insensitive search, price/percentage indicators, Chart.js Doughnut portfolio visualisation, and buy/sell actions using Material UI.

E-Commerce Microservices Platform · NestJS · PostgreSQL · Redis · Docker · Kubernetes

- Designed distributed NestJS microservices backend (products, orders, users, payments) with Redis caching, PostgreSQL connection pooling, and Kubernetes orchestration — Deployment/Service/ConfigMap manifests, horizontal pod autoscaling, and multi-stage Docker images with ClusterIP networking.

FraudGuard — Real-Time Fraud Detection System · Node.js · Flask · Next.js · Redis · Twilio API · Radiothon Hackathon · Top 10

- Led backend at a 24-hour hackathon — built Node.js REST API bridging Flask ML service to Next.js dashboard (sub-500ms latency); engineered 3-tier risk engine (Approve / Verify / Block) with Twilio SMS OTP; implemented Redis rate-limiting and request queuing for high-throughput reliability.

TripNest — Full-Stack Travel Accommodation Platform · Node.js · Express.js · MongoDB · Passport.js · Mapbox Live · Render + MongoDB Atlas

- Built RESTful MVC backend with Node.js, Express.js, and Mongoose; implemented Passport.js session-based authentication, role-based authorisation, and Helmet.js CSP hardening; integrated Mapbox Geocoding API for location-based search; deployed on Render + MongoDB Atlas

EDUCATION

VIT-AP University — B.Tech, Computer Science 2024 – 2028 · CGPA 7.78 / 10.0

ACHIEVEMENTS

- Top 10 at Radiothon Hackathon (100+ teams) — sole backend engineer for FraudGuard built end-to-end in 24 hours; completed two FlyRank AI production capstones (Billing Engine; DevPulse with LLM + Docker) and shipped Stockify and TripNest — delivering full-stack systems across the entire backend lifecycle